

Task 1: Conceptual Understanding

1. What is the difference between "Love" and "love" in NLP?

Ans: In NLP, "**Love**" and "**love**" are treated as **different tokens** because text is case-sensitive by default.

- "**Love**" → may represent a proper noun or start of a sentence
- "**love**" → commonly used as a verb or noun

If not normalized:

- The model treats them as separate words
- This increases vocabulary size and reduces efficiency

Therefore, we usually convert all text to lowercase so both become "**love**", ensuring consistency and better model performance.

2. What happens if stopwords are not removed?

Ans: If stopwords are not removed, the text will contain many **common, low-meaning words** like "*is*", "*the*", "*and*", which can negatively affect NLP models.

- **Increased noise** → Important words get diluted by unnecessary ones
- **Lower model accuracy** → Model may focus on irrelevant words
- **Larger vocabulary size** → More memory and computation required
- **Slower processing** → Training and prediction take more time

Overall, not removing stopwords can make the model **less efficient and less accurate**.

3. Provide two real-world scenarios where removing stopwords can be harmful.

Ans: 1. Sentiment Analysis

- Example: "*This is not good*"
- If the stopword "**not**" is removed → "*This is good*" ❌

This completely **changes the sentiment** from negative to positive. So, words like "**not**", "**no**" are crucial and should not be removed.

2. Chatbots / Conversational Systems

- Example: "*How are you?*"
- Removing stopwords → "?" or meaningless tokens

Stopwords help maintain **sentence structure and intent**, which is essential for understanding user queries.

Conclusion:

Removing stopwords blindly can lead to **loss of meaning and incorrect interpretations**, especially in context-sensitive applications.

4. Explain the difference between stemming and lemmatization.

Ans:

Feature	Stemming	Lemmatization
Definition	Cuts words to root form	Converts to meaningful base form
Approach	Rule - based	Dictionary + grammar
Output	May not be valid word	Always meaningful
Accuracy	Less accurate	More accurate
Speed	Faster	Slower
Example	studies → studi	studies → study
Use Case	When speed is important	When accuracy and meaning are important